

# What Must Be Trained and What Can Be Prompted: A Per-Capability Study of Tutor Redirect Behavior

Miles Yung

Independent Research

milesyung2026@gmail.com

## Abstract

Deploying a small language model as an English tutor forces a question a flat “good-dialogue” corpus never does: *which* tutor behaviors can be elicited by an explicit system prompt, and which must be demonstrated through fine-tuning? We answer this per-capability under a **matched-prompt** protocol — the identical fully-specified prompt given to a fine-tuned 0.8B student and to prompt-only baselines up to a 9B teacher. Within a single model family (Qwen) a sharp boundary emerges. Behaviors one clause elicits — locale fidelity, role-swap deflection, topic re-anchoring — reach prompt-only parity. Behaviors the prompt *describes but cannot install* do not: on persistence (firing a session-end sentinel on the third same-axis violation) the 9B teacher reaches  $\leq 0.06$  recall zero/few-shot and only 0.63 with native chain-of-thought (below the student’s 0.83); on pedagogical withholding, prompt-only models manage 0.09–0.45 versus the student’s 0.61. A cross-family probe (Llama-3.1-8B) corroborates the boundary’s direction. All experiments run on one RTX 3060 12GB GPU.

## 1 Introduction

**Problem.** We ask, for each behavior a deployed English tutor needs, whether a sufficiently explicit system prompt is enough or the behavior must be demonstrated in fine-tuning. A deployed tutor is governed by a long system prompt that already *states* most of what it should do — stay in character, ground culture in the learner’s locale, refuse role swaps, give one short example rather than a grammar lecture, escalate to a session-end signal under sustained abuse. If stating a behavior were sufficient, the data-side problem would be trivial.

**Why it matters.** It is not: the behaviors split cleanly into ones a prompt clause elicits and ones it only describes, and knowing which is which is exactly the decision a practitioner faces when building a task-specific model on consumer hardware —

spend the data budget only where a prompt will not do. The tutor setting makes the question answerable because each target behavior corresponds to a clause already present in a realistic deployment prompt (§3): the language of instruction, lesson topic, role structure, persona, pedagogical contract, locale frame, and general appropriateness.

**Gap.** The prompting-versus-fine-tuning trade-off has been studied at the level of *general* alignment — that a small demonstration set installs broad instruction-following (Zhou et al., 2023; Ouyang et al., 2022), and that in-context demonstrations often convey format more than new capability (Min et al., 2022) — but not resolved *per behavior* for a deployed task model. Synthetic-instruction pipelines and tutor-LLM datasets (§2) target general instruction-following with a uniform recipe; none asks, per behavior, whether the deployment prompt already elicits what the data teaches.

**Approach and contributions.** We supply the same fully-specified deployment prompt at evaluation to a fine-tuned 0.8B student and to a ladder of prompt-only baselines (0.8B base, 0.8B instruct, 4B instruct, and the 9B teacher that generated the training data; Figure~2). Because the instruction is present for every condition, a prompt-only failure cannot be blamed on under-specification. Our contributions:

1. **A per-capability map of the train-versus-prompt boundary** (Figure~3): promptable when a single clause both *describes* and *elicits* the behavior (locale fidelity, role-swap and topic re-anchoring), not promptable when it requires cross-turn state-tracking (persistence) or the suppression of a strong competing prior (pedagogical withholding).
2. **An evidenced negative result about the conventional metric:** per-axis redirect F1 is a context-blind *type* classifier that ties a 0.8B student with the 9B teacher; we replace it with

matched per-capability instruments (§5).

3. **Reusable apparatus:** a locale-aware, yield-aware generation pipeline (§3) and a *partially validated* trigger-position decorrelation construction for the persistence sentinel (§5).

All experiments run on one RTX 3060 12GB GPU — the regime where the boundary matters most, since a large general-purpose model is not deployable there. We evaluate one family (Qwen; a genuine limitation, §6) and one locale (which does not limit the persistence/withholding boundary, both structural behaviors independent of the locale backdrop).

## 2 Related Work

**Synthetic instruction data.** Teacher-to-student bootstrapping was popularised by Self-Instruct (Wang et al., 2023) and Evol-Instruct / WizardLM (Xu et al., 2024); UltraChat (Ding et al., 2023) extends to multi-turn dialogue, and Tulu-3 (Lambert et al., 2024) composes datasets from sub-pools. These target *general* instruction-following with a uniform recipe and do not ask which taught behaviors a prompt already elicits.

**LLM-as-judge filtering.** Prometheus (Kim et al., 2024), JudgeLM (Zhu et al., 2023), and PandaLM (Wang et al., 2024) propose dedicated judges; a strand documents position, length, and self-preference bias (Saito et al., 2023; Panickssery et al., 2024), mitigated by multi-judge consensus. Our pairwise eval (§5) follows this with a cross-family constraint so no judge shares the teacher’s lineage; we add a *negative* result about a judged metric (context-blind redirect F1).

**Tutor and educational LLMs.** EduChat (Dan et al., 2023) and MathDial (Macina et al., 2023) target educational dialogue; CEFR-aligned corpora (EFCAMDAT (Geertzen et al., 2014), TLE (Berzak et al., 2016)) are *learner-produced*, not tutor-side. None asks, per behavior, whether the deployment prompt already suffices. MathDial’s scaffolding moves are the closest analogue to our pedagogical-withholding axis.

**Safety, persona, and shortcut learning.** Adversarial safety data (Parrish et al., 2022; Mazeika et al., 2024; Zou et al., 2023) and persona consistency (Zhang et al., 2018; Deshpande et al., 2023) mostly treat redirection as *single-turn*. Our three-strike persistence is multi-turn and carries a *shortcut-learning* (Geirhos et al., 2020; McCoy et al., 2019) risk: a positionally-regular sentinel

| Invariant / violation             | Minimal repair                             |
|-----------------------------------|--------------------------------------------|
| Language (code-switch to L1)      | Acknowledge, steer back to target language |
| Topic (off-topic drift)           | Re-anchor to the subject                   |
| Role (“you be the learner”)       | Decline, reassert tutoring structure       |
| Persona (“are you a chatbot?”)    | Reassert the frame, continue in persona    |
| Pedagogy (“just give the answer”) | Scaffold rather than supply the answer     |
| Locale (out-of-locale reference)  | Brief in-locale redirect                   |
| Appropriateness (politics/etc.)   | Generic safe redirect (catch-all)          |

Table 1: **The seven single-shot redirect axes** as invariant/violation/minimal-repair triples, read off the deployment prompt.

invites a position-as-trigger shortcut, which our decorrelation construction (§3) targets.

**Positioning.** The prompting-vs-fine-tuning question has been examined for *general* alignment (Zhou et al., 2023; Ouyang et al., 2022; Min et al., 2022); we localize it to the *per-behavior* level for a deployed task model, under a matched-prompt protocol giving the same prompt to trained and prompt-only models alike.

## 3 Method

**Pipeline.** A single locally-served 9B teacher generates all training data from a pool of CEFR-stratified scenario seeds through twelve parallel SFT streams, a six-filter cascade, and a yield-aware top-up loop (Figure~1). The teacher receives a prompt of exactly the shape the student sees at deployment, so there is no distribution shift between demonstration and inference. Full pipeline mechanics (top-up equations, filter cascade, hash-deterministic split) are in the appendix and the released code.

**An invariant-based taxonomy.** We read the redirect axes off the deployment prompt rather than intuiting them. A tutor is the keeper of a set of **interaction invariants**, each corresponding to one commitment the prompt makes; a learner *violation* breaks exactly one, and the correct redirect is the **minimal repair** that restores it (Table~1). This makes the axis list auditable (one clause one axis) and converts “a generic redirect stream is insufficient” into a per-axis falsifiable prediction: specialized data should help exactly on axes whose repair is not already supplied by assistant priors or a prompt clause (§5).

## Locale-aware, yield-aware generation pipeline (all on one RTX 3060 12GB)

*single locally-served 9B teacher generates all streams*

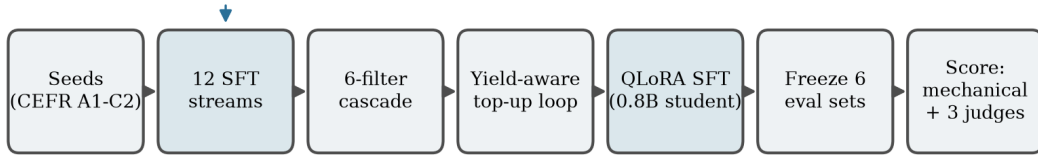


Figure 1: **The locale-aware, yield-aware generation pipeline.** A single locally-served 9B teacher drives twelve SFT streams; a six-filter cascade and a yield-aware top-up loop shape the corpus, which trains a 0.8B QLoRA student scored on six frozen held-out sets — all on one RTX 3060 12GB.

The pipeline realises this through twelve streams: *normal* scaffolded dialogue (1), *single-shot redirects* (7, one per axis), and *persistent 3-strike* streams (4, for the axes where repeated violation warrants escalation to a session-end sentinel).

**Deployment prompt.** Every axis appears as an explicit [guidelines] clause, and a [persistence] block specifies the full three-strike protocol: first warm redirect, second shorter redirect, third a warm sentence plus the exact literal sentinel [SESSION\_END: <axis>]. This is what makes the boundary interpretable — when a prompt-only model fails on persistence or withholding (§5), it fails *despite* the behavior being spelled out.

**Trigger-position decorrelation.** A fixed-turn sentinel makes turn position a perfect proxy for “third strike,” inviting a positional shortcut. We resample the sentinel turn over a parity-constrained set {5, 7, 9, 11} (hash-deterministic per seed; every variant holds the trigger at exactly three strikes, varying only lead-in scaffolding). We report this as *partially validated* (§5).

## 4 Experimental Setup

**Models and recipe.** Student base Qwen3.5-0.8B-Base; teacher Qwen3.5-9B-UD-Q4\_K\_XL (4-bit, llama.cpp), which also serves as the strongest prompt-only baseline (B4). All experiments on one RTX 3060 12GB. Every trained condition uses the *same* recipe — QLoRA (Detrmers et al., 2023) (NF4 4-bit) with LoRA (Hu et al., 2022) rank 16 / alpha 32, 2 epochs, no DPO — so each ablation is single-variable (hyperparameters in the appendix).

**Conditions.** **A1** — the student, full 12-stream mix. **A3** — A1 minus the six specialized single-shot redirect streams (the *generic-SFT* baseline). **A5** — A1 with persistence rebuilt fixed-turn-7 (the decorrelation contrast). Prompt-only: **B1** 0.8B base, **B2** 0.8B instruct, **B3** 4B instruct, **B4** 9B teacher. All see the identical deployment prompt (Figure~2).

**Held-out sets** (frozen, locale=china): Tutor-Scenario (224), Redirect-Probe (143), and four persistence probes isolating recall (Persistent-Probe, 159 positives) from two false-positive channels (Persistent-FP-Probe benign negatives at trained positions; Persistent-Premature-Probe under-threshold negatives) and off-grid firing (Persistent-OffPosition-Probe).

**Metrics.** Persistence and locale leakage are **mechanical** (judge-free): sentinel-firing recall against literal [SESSION\_END: <axis>] strings, and Western-default leakage via a word-boundary gazetteer. Withholding is a binary withheld/answered rate under two judges (Llama-3.1-8B, Gemma-2-9B). Repair *quality* on the promptable axes is a pairwise preference under a cross-family three-judge ensemble (Prometheus / Llama-3.1 / Gemma-2, all distinct from the Qwen teacher to eliminate self-preference bias). The load-bearing A1/A3 conditions are reported over **three seeds** (42/123/7); others single-seed.

## 5 Results: The Train-versus-Prompt Boundary

Every condition is evaluated under the **same** fully-specified deployment prompt (§3), so a prompt-only failure isolates *promptability* rather than under-

Matched-prompt protocol: one prompt, a ladder of conditions, one instrument per capability

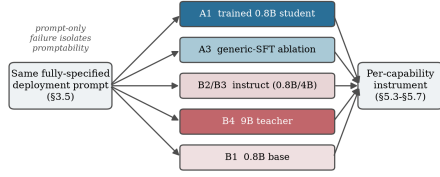


Figure 2: **Matched-prompt protocol.** The identical deployment prompt is given to a fine-tuned 0.8B student, a generic-SFT ablation, and prompt-only baselines up to the 9B teacher; each is scored by the instrument matched to the capability.

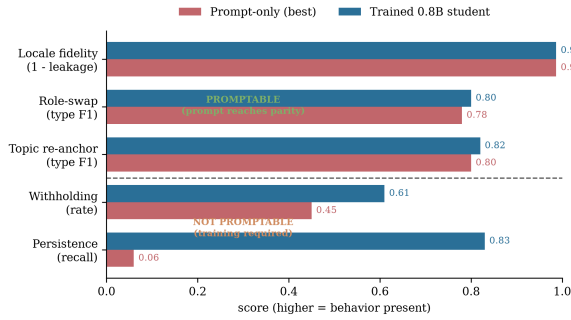


Figure 3: **The boundary.** *Top:* behaviors one clause elicits — prompt-only reaches parity. *Bottom:* behaviors the prompt describes but cannot install — prompt-only falls far short of the trained 0.8B student. Values from Table 2.

specification (Figure~2). The behaviors separate sharply (Figure~3, Table~2).

### 5.1 Not promptable I — multi-turn persistence

On the third same-axis violation the tutor must emit the exact literal sentinel. Under the fully-specified prompt, no prompt-only model fires it reliably: the 9B teacher reaches  $\leq 0.06$  recall zero/few-shot, and the no-data ablation A3 fires on **0.000** of positives, while every model trained on the persistent streams fires (A1 recall **0.83**, 3-seed mean  $0.85 \pm 0.04$ ).

**The gap is not an artifact of zero-shot prompting.** We ran a prompting ladder on the strongest prompt-only model (the 9B teacher; few-shot exemplars span positions {5,7,9,11} so they cannot teach a fixed turn). Table~3 and Figure~4 (left) show few-shot and an output counting-scaffold do *not* help ( $\leq 0.06$ ); only Qwen3.5’s *native* chain-of-thought moves recall, to 0.63 — still below the trained 0.83, and at 1.6–3.2k reasoning tokens/turn with frequent delivery failures (the fire decision is reached inside <think> but omitted from the

| Capability (metric)  | Prompt-only            | Trained     |
|----------------------|------------------------|-------------|
| Persistence (recall) | $\leq 0.06$ / A3 0.000 | <b>0.83</b> |
| Withholding (rate)   | 0.09–0.45              | <b>0.61</b> |
| Locale (1 – leakage) | 0.987                  | 0.987       |
| Role-swap (type F1)  | 0.71–0.86              | 0.75–0.86   |
| Topic (type F1)      | saturated              | saturated   |

Table 2: **The boundary.** *Top:* not-promptable (the 9B teacher and, for persistence, the no-data ablation A3 fail despite the explicit instruction). *Bottom:* promptable (prompt-only parity). Persistence prompt-only is the 9B teacher’s zero/few-shot recall; native CoT reaches 0.63 (§5.1). Withholding prompt-only spans B1–B4 (9B teacher 0.45).

| 9B teacher prompting condition   | recall      |
|----------------------------------|-------------|
| zero-shot instruction            | 0.025       |
| + few-shot exemplars             | 0.013       |
| + CoT output scaffold (no-think) | 0.057       |
| + native chain-of-thought        | <b>0.63</b> |
| A1 trained 0.8B (reference)      | <b>0.83</b> |

Table 3: **Persistence resists prompting up to, but not including, native CoT — and even that stays below the trained student.** Recall on Persistent-Probe positives (n=159).

visible answer). Persistence is thus a *relocated* boundary: it resists zero/few-shot outright and is only partially recovered by native CoT, at an accuracy, reliability, and inference-cost deficit SFT removes.

**Decorrelation, partially validated.** In an isolated matched contrast (A1 4-variant vs A5 fixed-turn-7, 1-epoch budget), decorrelation *removes the positional recall bias* — the fixed-turn design recalls best at its trained turn 7 and degrades off-position (overall 0.56), while the 4-variant fires uniformly across positions (0.82) — but does *not* reduce premature firing (0.208 vs 0.119). At 0.8B the premature over-firing tracks accumulated violation count and conversation depth (peaking at turn 9, not the trained 7), so it is threshold-laxity, not a positional shortcut for decorrelation to suppress.

### 5.2 Not promptable II — pedagogical withholding

The tutor must *withhold* the requested answer and scaffold instead, an instruction the prompt states explicitly. We score 63 held-out probes under two judges (Table in Figure~4, right). The trained stu-



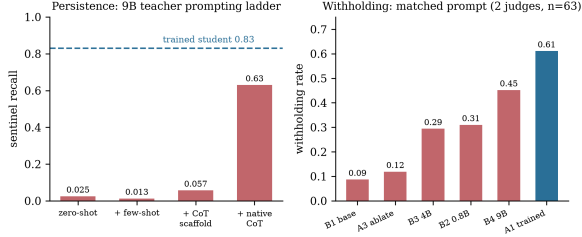


Figure 4: **The two not-promptable behaviors.** *Left:* on persistence, the 9B teacher stays at  $\leq 0.06$  through few-shot and an output CoT scaffold; only native CoT moves it, to 0.63 — still below the trained 0.83 (dashed). *Right:* on withholding, every prompt-only condition (incl. the 9B teacher, 0.45) falls below the trained 0.61, and the no-data ablation A3 collapses to baseline.

dent withholds at **0.61**; the no-specialized-data ablation A3 collapses to 0.12, near the untrained-base rate — clean evidence the behavior is *installed by demonstration* (A1-vs-A3 clears significance under each judge,  $z = 5.9 / 5.6$ ,  $p \ll 0.001$ ). Every prompt-only model, including the 9B teacher (0.45), withholds less than the trained 0.8B student despite the identical instruction: a model an order of magnitude larger, told plainly to scaffold, complies on under half of probes. (The student-beats-teacher *comparison* is directional only at  $n=63$ ; the boundary rests on the teacher’s absolute non-compliance and the A3 ablation.) Scaffolding requires suppressing the model’s strong helpfulness prior — a clause can describe that suppression but not produce it.

### 5.3 Promptable axes — type is prompted, quality is refined

Redirect-axis F1 from a context-blind judge is **uninformative**: it is a *type* classifier that floors on axes with no context-free surface form (locale, language, topic) and ceilings on axes every model satisfies (persona, role-swap), tying A1, B2, and the 9B teacher at 0.409. We therefore score *quality* on the promptable axes with a pairwise preference (A1 vs A3, both SFT-only, differing only in the specialized streams). Every axis favors A1; the largest effect is **role-swap (win-rate 0.87)** — on an axis F1 called saturated parity, so the F1 ceiling concealed a real, large specialized-data effect. A generic-redirect negative control (both conditions train on it) sits at 0.55, so the wins are axis-specific, not a global “A1 is better” artifact. **Locale fidelity is prompt-driven**: Western-default leakage is statistically indistinguishable across trained and prompt-only models (A1 1.34%, A3 0.89%,

B1 1.34%), and ablating the locale stream does not increase leakage — it is carried by one prompt clause.

## 6 Discussion and Conclusion

Under a matched-prompt protocol, tutor redirect behaviors separate along an interpretable line: **promptable** when a single clause both *describes* and *elicits* the behavior, and **not promptable** when it requires cross-turn state-tracking (persistence) or the suppression of a strong competing prior (withholding) — things a clause can name but not produce. The boundary tells a deployer of a small model which behaviors a prompt gives for free (locale, role-swap, topic) and which require the pipeline (persistence, withholding). Reaching it cleanly required retiring the conventional context-blind redirect-axis F1 — a *type* classifier that ties a 0.8B student with a 9B teacher — for matched per-capability instruments, which recover the largest specialized-data effect in the study (role-swap quality, win-rate 0.87) on an axis F1 called saturated parity.

**Threats to validity.** *Single family* (Qwen) is the load-bearing generalization gap: whether a behavior is trainable-but-not-promptable could shift with a family’s instruction-following strength. A cross-family *prompt-only* probe on **Llama-3.1-8B-Instruct** under the matched prompt confirms the boundary’s *direction* holds outside Qwen (withholding 0.22–0.32 vs trained 0.63; persistence recall 0.27 zero-shot, 0.55 with prompted CoT, vs trained 0.85), though the *magnitude* is family-dependent (Llama attains more prompt-only persistence than the Qwen teacher). A *trained* non-Qwen student is the definitive test and is left to future work. *Single locale* does not threaten the central boundary: persistence and withholding are structural behaviors independent of the locale backdrop. *Judged-metric power*: the trained-vs-teacher withholding gap is directional at  $n=63$ ; the strong claim is the teacher’s absolute sub-50% compliance. *Decorrelation* is partially validated (fixes positional recall, not premature firing).

**Future work**, in priority order: (i) a **trained cross-family student** (the load-bearing test, since family, not locale, is where the boundary could move); (ii) **larger-scale decorrelation**, where the positional route is cheaper relative to the semantic one; (iii) a **better-powered pedagogy teacher comparison**; and (iv) transfer of the matched-prompt methodology to other rare, semantically-

triggered markers (refusal-token and tool-call emission), where the same “describable but not promptable” question applies.

**Conclusion.** Which tutor behaviors must be trained and which can be prompted is answerable per capability, and the answer is sharp: a prompt clause installs a behavior only when it both describes *and* elicits it. Persistence and pedagogical withholding are described by the prompt but installed only by demonstration — the central finding, established on a single consumer GPU where the distinction matters most.

## 6.1 Ethics and data statement

All training and evaluation data are model-generated (distilled from a locally-served teacher) and contain no personal data or PII; no human subjects were involved. The intended use is research on data curation and the train-versus-prompt boundary for small task-specific models.

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## A Reproducibility and Hyperparameters

**Training.** QLoRA (Dettmers et al., 2023) (NF4 4-bit, double-quant, bfloat16) with LoRA (Hu et al., 2022) rank 16, alpha 32, dropout 0.05 over the seven linear projections of the language tower; 2 epochs, peak LR  $2 \times 10^{-4}$  cosine, effective batch 8, max sequence length 1792, paged\_adamw\_8bit, SDPA attention. No DPO enters any condition. Teacher served via llama.cpp (context 32 768, 80 GPU layers); teacher and trainer are swapped since they cannot co-reside on 12 GB.

**Persistent-variant codomain.** Sentinel positions {5,7,9,11} are forced: dialogues open on a learner turn (only odd positions valid), with  $\geq 5$  to fit three strikes and two probes and  $\leq 11$  to stay under the 1792-token cap. Variant assignment `int(sha256(seed_id)[:8]) % 4` is seeded (not random) so re-generation is stable and the train/eval split stays coherent.

**Pipeline filters.** Six filters run cheap-to-expensive: five mechanical (script, banned terms, schema, naturalness) then one LLM-judge (locale\_judge). An audit of the locale\_judge found ~85% false positives on its rejections (common English sentence-initial words, locally-canonical landmarks), remediable with static allowlists (global pass rate 70.1%  $\rightarrow$  88.4%) — a reusable caution for any capitalization-based entity filter.

**Artifacts.** One Python codebase; generation driven by config/generation.yaml, train-

ing by per-condition YAMLS, the six held-out sets frozen by `scripts/build_eval_sets.py`. The full-length version of this paper reports the per-capability statistics, the F1-bimodality negative result, and the complete confound analysis.